

# SCFDepth: A Single-step Coarse-to-Fine Diffusion Framework for Monocular Depth Estimation

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**Abstract**—Diffusion-based monocular depth estimation (MDE) has shown strong zero-shot generalization by transferring rich visual priors from large-scale generative models. However, its iterative denoising process requires repeated U-Net evaluations, making inference inefficient for practical deployment. Recent single-step diffusion methods improve efficiency by replacing the stochastic reverse process with deterministic prediction, but the compressed inference often weakens global structure and blurs fine-grained depth discontinuities. To address this efficiency–fidelity trade-off, we propose SCFDepth, a single-step coarse-to-fine diffusion framework for fast and detail-preserving monocular depth estimation. Instead of relying on RGB latents alone, SCFDepth introduces the prediction of a pretrained feed-forward MDE model as a semantic structural prior, which serves as an auxiliary latent condition to guide scene-level depth generation. To further adapt the diffusion backbone to geometric perception, we develop a two-stage training strategy. First, intermediate U-Net features are aligned with representations from a frozen vision encoder through a spatial-preserving convolutional adapter and cosine similarity loss, enhancing semantic structure awareness. Second, the RGB latent and semantic-prior latent are jointly used for single-step depth prediction, where spatial-domain supervision establishes coherent global geometry and a frequency-domain objective refines high-frequency boundaries. Extensive zero-shot experiments on five benchmarks demonstrate that SCFDepth achieves competitive performance among diffusion-based methods while requiring only one denoising step. Code is available at <https://dimon000000.github.io/SCFDepth/>.

**Index Terms**—Diffusion Model, Depth Estimation, Two-Stage Training, Frequency Domain

## I. INTRODUCTION

**M**ONOCULAR depth estimation (MDE) predicts a dense depth map from a single RGB image and is fundamental to autonomous driving, robotics, and augmented reality. Since the projection from a three-dimensional scene to a two-dimensional image discards depth information, multiple 3D structures may produce similar image observations, making monocular depth estimation inherently ill-posed. Therefore, an effective model must capture both global scene structure and fine-grained depth discontinuities. Recent feed-forward methods, such as Depth Anything V2 [1] and ScaleDepth [2], achieve strong zero-shot generalization through large-scale

training. In parallel, diffusion-based methods exploit pre-trained generative priors and have demonstrated competitive geometric accuracy with considerably fewer depth annotations [3], [4].

However, the iterative denoising process of diffusion models requires repeated evaluations of a computationally expensive U-Net, leading to high inference latency. Recent approaches, including GenPercept [5] and Lotus [6], compress the reverse process into a deterministic single step. Although this significantly improves efficiency, removing iterative refinement often produces over-smoothed depth maps, blurred object boundaries, and incomplete thin structures.

The difficulty lies in requiring a single forward pass to recover semantic context, global geometry, and high-frequency details simultaneously. Moreover, diffusion backbones are originally pretrained for appearance reconstruction and may preserve texture variations unrelated to geometric changes. Without progressive denoising, these irrelevant cues cannot be gradually suppressed, while local depth discontinuities receive insufficient refinement. Therefore, an effective single-step framework requires both reliable geometric guidance and a progressive optimization strategy.

To address this efficiency–fidelity trade-off, we propose **SCFDepth**, a coarse-to-fine single-step diffusion framework. A pretrained feed-forward MDE model first provides an explicit global depth prior, reducing the burden of inferring the entire scene geometry from RGB features alone. SCFDepth then employs a two-stage training strategy. In Stage 1, intermediate U-Net features are aligned with DINOv2 representations through a spatial-preserving convolutional adapter and cosine similarity loss, strengthening structural understanding. In Stage 2, latent- and pixel-domain losses first establish stable global depth, after which a weighted Fourier-domain objective refines high-frequency boundaries and local details. The final model performs deterministic latent-to-latent prediction with only one U-Net evaluation.

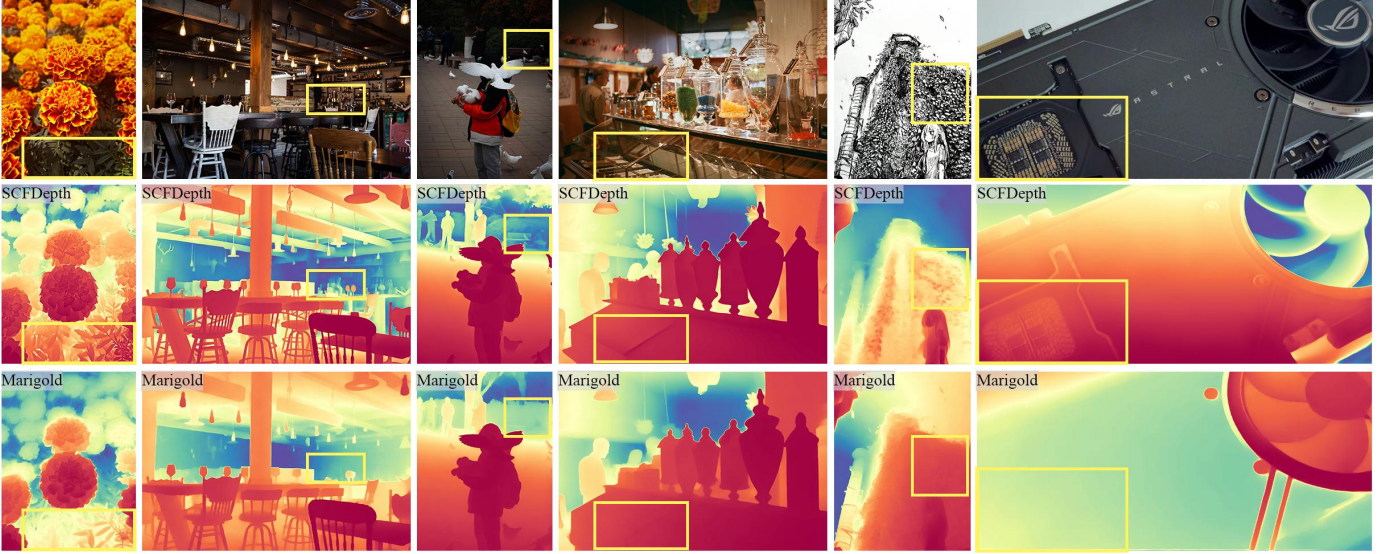
Extensive experiments on standard zero-shot monocular depth estimation benchmarks demonstrate the superiority of our approach. Compared with state-of-the-art models, **SCFDepth** achieves highly competitive geometric accuracy and preserves sharp edge details, all with significantly reduced inference time. To summarize, our contributions are:

- **A Single-Step Diffusion Framework:** We propose a deterministic single-step paradigm for diffusion-based depth estimation that incorporates a pretrained data-driven MDE model as a geometric prior, significantly accelerating inference while preserving robust global depth structures.

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- **Enhanced Feature Alignment via Cosine Similarity:** By introducing a spatial-preserving Conv Adapter and a cosine similarity loss to explicitly align the intermediate U-Net features with an external encoder, we substantially enrich the model’s structural understanding and mitigate texture overfitting.
- **Spatial- and Frequency-Domain Loss Strategy:** We refine the second training stage into a coarse-to-fine learning process. It first employs latent MSE and pixel  $L_1$  losses for global metric accuracy, followed by a transition to a novel frequency-domain (FFT) loss to explicitly supervise high-frequency components and recover sharp local edge details.

## II. RELATED WORK

### A. Monocular Depth Estimation

Early learning-based MDE methods formulated depth prediction as direct regression using multi-scale CNNs [7]. Subsequent studies improved depth representation and structural modeling through ordinal regression [8], local planar guidance [9], adaptive depth bins [10], attention-based decoding [11], and neural CRFs [12]. Vision transformers further enhanced long-range dependency modeling by integrating multi-level features for dense prediction [13]. Although these architectures substantially improved in-domain accuracy, their cross-dataset performance remained sensitive to the diversity and depth distribution of the training data.

To improve generalization, MiDaS adopted mixed-dataset training and affine-invariant objectives for transferable relative depth estimation [14], while LeReS recovered scene geometry through image-level scale and shift estimation [15]. Metric3D and Metric3D V2 incorporated camera information for zero-shot metric depth prediction [16], [17]. Depth Anything and Depth Anything V2 further scaled pseudo-labeled training with DINOv2 representations and large image collections [1], [18]. These feed-forward models provide strong semantic representations and efficient inference. SCFDepth uses such a model as a global geometric prior rather than the final

predictor, allowing the diffusion branch to refine ambiguous regions and depth boundaries.

### B. Diffusion Models

Diffusion models originated from nonequilibrium thermodynamics [19] and were made practical by DDPMs [20]. Score-based modeling and stochastic differential equation formulations later provided a unified interpretation of the denoising process [21]–[23]. These models have demonstrated strong generative priors in image synthesis [24], [25], inpainting [26], [27], and super-resolution [28], [29]. DDPMs define a forward Markov chain that gradually perturbs a clean latent  $\mathbf{z}_0$  with Gaussian noise. Given a variance schedule  $\beta_t$ , one transition is

$$q(\mathbf{z}_t|\mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{z}_t; \sqrt{1 - \beta_t}\mathbf{z}_{t-1}, \beta_t\mathbf{I}). \quad (1)$$

The latent at an arbitrary timestep can be sampled directly as

$$\mathbf{z}_t = \sqrt{\bar{\alpha}_t}\mathbf{z}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \quad (2)$$

where  $\alpha_t = 1 - \beta_t$ ,  $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ , and  $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ . The learned reverse transition is

$$p_\theta(\mathbf{z}_{t-1}|\mathbf{z}_t) = \mathcal{N}(\mathbf{z}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{z}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{z}_t, t)). \quad (3)$$

A commonly used reweighted objective predicts the injected noise:

$$\mathcal{L}_{\text{simple}}(\theta) = \mathbb{E}_{t, \mathbf{z}_0, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{z}_t, t)\|_2^2]. \quad (4)$$

Pixel-space diffusion is computationally expensive because the reverse process repeatedly operates on high-resolution tensors. Latent Diffusion Models reduce this cost by performing denoising in a compressed VAE space and exploiting priors learned from large-scale image–text datasets such as LAION-5B [24], [30]. Sampling efficiency has also been improved through DDIM, noise estimation, shortened schedules, cascaded models, and latent score modeling [31]–[35], while flow matching provides an alternative continuous-time formulation [36]. Nevertheless, reducing the number of steps does not eliminate the mismatch between appearance reconstruction and geometric perception, motivating semantic representation alignment [37]–[39].



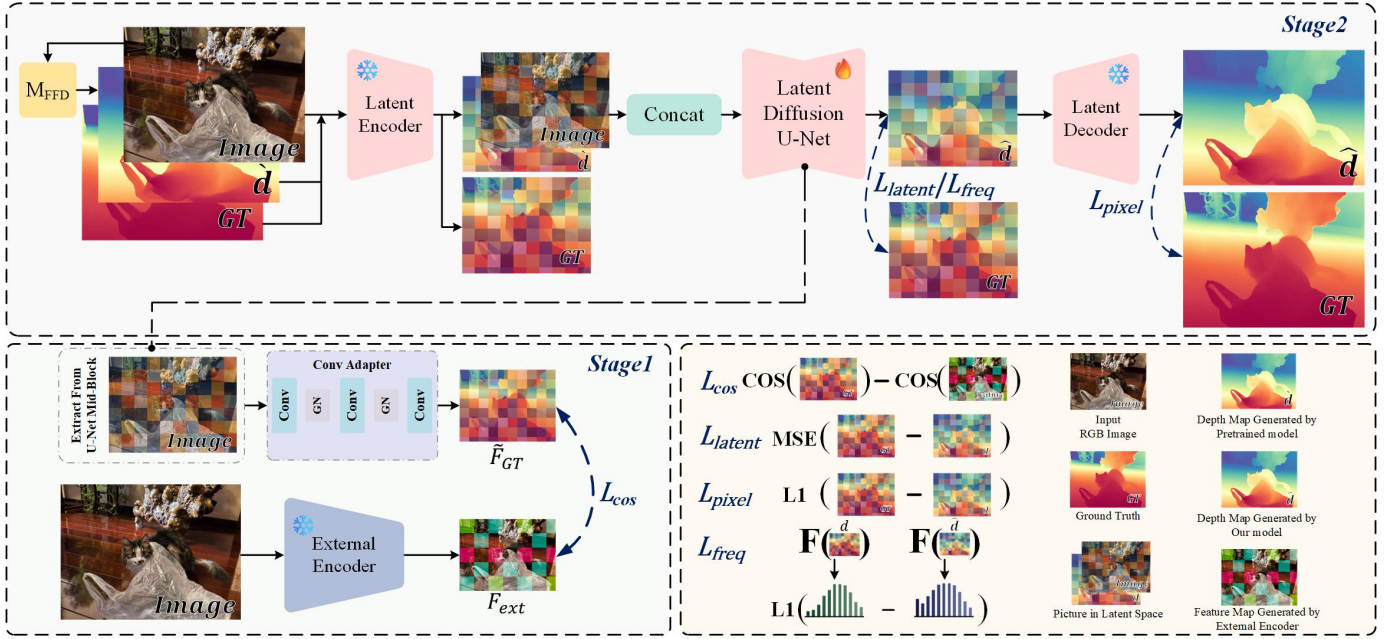


Fig. 1. **Overview of the SCFDepth two-stage training framework.** **Stage 1 (Bottom Left):** To establish a robust semantic foundation, the intermediate features of the U-Net are processed by a spatial-preserving Conv Adapter and explicitly aligned with representations from a frozen external encoder via a cosine similarity loss ( $L_{cos}$ ). **Stage 2 (Top):** For fine-grained depth estimation, the RGB image and a semantic prior generated by a pre-trained  $M_{FFD}$  model are encoded into the latent space and concatenated. The trainable Latent Diffusion U-Net predicts the depth latent, supervised by an iteration-based loss scheduling strategy. Initially, spatial constraints ( $L_{latent}$  and  $L_{pixel}$ ) enforce global metric accuracy. In the final phase, a frequency-domain loss ( $L_{freq}$ ) is introduced to reconstruct sharp, high-frequency edge details.

### C. Diffusion-based Depth Estimation

Recent studies have reformulated MDE as conditional generation and demonstrated the robustness of diffusion priors under challenging visual conditions [40]. DDP and DiffusionDepth developed iterative image- or latent-space denoising pipelines [3], [41], DepthGen combined supervised and self-supervised objectives [42], and Marigold repurposed a pretrained latent diffusion model for zero-shot depth estimation [4]. E2E-FT and GenPercept simplified task adaptation and direct prediction [5], [43], while DepthFM and Lotus reduced inference cost through flow matching or single-step generation [6], [44]. However, compressing the reverse process weakens progressive correction and often degrades thin structures and depth discontinuities. DepthMaster addressed this limitation through staged representation alignment and Fourier enhancement [45]. SCFDepth retains deterministic single-step inference while introducing an explicit feed-forward depth prior, a spatial-preserving adapter, and a scheduled spatial-to-frequency objective.

## III. METHOD

### A. Single-Step Formulation and Overview

A conventional conditional latent diffusion model starts from a Gaussian latent  $\mathbf{z}_T^{(y)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  and progressively reconstructs the target through a reverse denoising chain. At each timestep, the U-Net predicts the noise conditioned on the input image latent  $\mathbf{z}^{(x)}$ . A representative stochastic update is

$$\mathbf{z}_{t-1}^{(y)} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{z}_t^{(y)} - \frac{1 - \alpha_t}{\sqrt{1 - \alpha_t}} \epsilon_\theta(\mathbf{z}_t^{(y)}, \mathbf{z}^{(x)}, t) \right) + \sigma_t \mathbf{w}. \quad (5)$$

The iterative process gradually suppresses irrelevant appearance information and refines the target structure. However, repeatedly evaluating the U-Net makes inference cost approximately proportional to the number of denoising steps.

SCFDepth replaces this stochastic reverse chain with a deterministic single-step mapping. The timestep is fixed to  $t = 1$ , and the U-Net directly predicts the target depth latent from the encoded conditions:

$$\mathbf{z}_{pred} = \mathcal{F}_\theta([\mathbf{z}^{(x)}, \mathbf{z}^{(p)}], 1). \quad (6)$$

Here,  $\mathbf{z}^{(x)}$  denotes the RGB latent and  $\mathbf{z}^{(p)}$  denotes the latent representation of the pretrained MDE prediction. The two representations are concatenated along the channel dimension and jointly processed by the U-Net. The predicted latent  $\mathbf{z}_{pred}$  is subsequently decoded by the VAE to obtain the depth map. This formulation removes the iterative reverse process and requires only one U-Net evaluation.

Although single-step prediction significantly improves efficiency, it also removes the progressive correction provided by multi-step denoising. The network must recover semantic structure, global depth arrangement, and local discontinuities within a single forward pass. Jointly optimizing these objectives can cause the model to overfit appearance textures or produce overly smooth depth boundaries. To alleviate this conflict, SCFDepth adopts a coarse-to-fine two-stage training strategy, as illustrated in Fig. 1. Stage 1 improves the structural representation of the diffusion U-Net through feature alignment, while Stage 2 introduces geometric guidance and progressively learns global depth and high-frequency details.

The separation of the two stages is motivated by the different optimization requirements of semantic representation and depth reconstruction. Direct depth supervision applied

before the U-Net has acquired reliable structural features may encourage dataset-specific texture fitting. Similarly, strong high-frequency supervision introduced before global geometry is established may amplify local gradients and destabilize the scene layout. SCFDepth therefore first regularizes the representation space, then learns coarse depth structure, and finally refines local boundaries.

### B. Stage 1: Spatial-Preserving Feature Alignment

Diffusion U-Nets are pretrained for image reconstruction and may retain appearance cues that are weakly related to scene geometry. To strengthen their structural representation, we align an intermediate U-Net feature  $F_{\text{unet}}$  with features  $F_{\text{ext}}$  extracted by a frozen DINOv2 encoder [46].

Because dense depth estimation depends on spatial correspondence, we introduce a ConvFeatureAdapter instead of a point-wise projection. The adapter preserves the two-dimensional feature layout while matching the channel dimension of the external representation:

$$F_1 = \sigma(\text{GN}(\text{Conv}_{1 \times 1}(F_{\text{unet}}))). \quad (7)$$

$$F_2 = \sigma(\text{GN}(\text{Conv}_{3 \times 3}(F_1))). \quad (8)$$

$$\tilde{F}_{\text{unet}} = \text{Conv}_{1 \times 1}(F_2). \quad (9)$$

The first  $1 \times 1$  convolution projects the channel dimension, the  $3 \times 3$  convolution aggregates local spatial context, and the final layer maps the feature to the external embedding dimension. Group normalization is adopted because it remains stable under the small batch sizes commonly used for diffusion training.

After resizing  $F_{\text{ext}}$  to the spatial resolution of  $\tilde{F}_{\text{unet}}$ , both representations are normalized along the channel dimension:

$$\hat{F}_{\text{unet}} = \frac{\tilde{F}_{\text{unet}}}{\|\tilde{F}_{\text{unet}}\|_2}, \quad \hat{F}_{\text{ext}} = \frac{F_{\text{ext}}}{\|F_{\text{ext}}\|_2}. \quad (10)$$

The feature-alignment objective is defined as

$$\mathcal{L}_{\text{fa}} = 1 - \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W \sum_{c=1}^C \hat{F}_{\text{unet}}^{(c,i,j)} \cdot \hat{F}_{\text{ext}}^{(c,i,j)} \quad (11)$$

Cosine similarity aligns semantic directions rather than absolute activation magnitudes. This avoids forcing features from the frozen encoder and the diffusion U-Net to share identical numerical scales, while encouraging corresponding spatial locations to encode consistent scene structure.

### C. Stage 2: Coarse-to-Fine Depth Refinement

After Stage 1 establishes a structurally informed U-Net representation, Stage 2 optimizes the network for depth prediction. As shown in Eq. (6), the RGB image and the prediction of Depth Anything V2 [1] are independently encoded into  $\mathbf{z}^{(x)}$  and  $\mathbf{z}^{(p)}$ . The RGB latent preserves image evidence, whereas the prior latent provides an initial estimate of object ordering, large surfaces, and long-range scene layout. Their concatenation allows the trainable U-Net to use the prior as geometric conditioning while correcting its errors according to the supervised ground-truth depth.

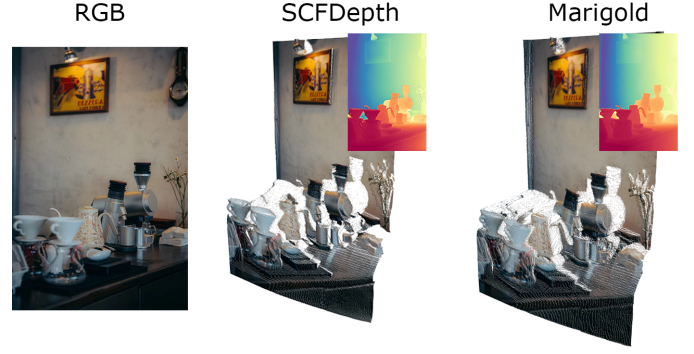


Fig. 2. Point-cloud comparison. Single-step SCFDepth produces fewer flying points and more coherent surfaces than multi-step Marigold in this example.

Stage 2 follows a coarse-to-fine optimization schedule. During the first 20K iterations, the network is jointly supervised in the latent and pixel domains:

$$\mathcal{L}_{\text{phase1}} = \text{MSE}(\mathbf{z}_{\text{pred}}, \mathbf{z}_{\text{gt}}) + \mathcal{L}_1(\hat{\mathbf{y}}, \mathbf{y}_{\text{gt}}). \quad (12)$$

The latent MSE constrains the prediction within the VAE representation space and provides stable supervision at the U-Net resolution. The pixel-level  $L_1$  loss directly penalizes decoded depth errors, ensuring consistency between the latent prediction and the target depth map. Together, these losses establish the global scene structure before detail refinement.

For the final 1K iterations, the spatial losses are replaced by a frequency-domain objective:

$$\mathcal{L}_{\text{phase2}} = \lambda_{\text{freq}} \mathcal{L}_{\text{freq}}. \quad (13)$$

Given the predicted depth map  $\hat{D}$  and ground-truth depth map  $D$ , their two-dimensional Fourier transforms are

$$\mathcal{F}_{\hat{D}} = \text{FFT2}(\hat{D}), \quad \mathcal{F}_D = \text{FFT2}(D). \quad (14)$$

The corresponding centered magnitude spectra are

$$M_{\hat{D}} = |\text{FFTShift}(\mathcal{F}_{\hat{D}})|, \quad M_D = |\text{FFTShift}(\mathcal{F}_D)|. \quad (15)$$

Their unweighted spectral discrepancy is defined as

$$\mathcal{L}_{\text{base}} = \|M_{\hat{D}} - M_D\|_p, \quad p \in \{1, 2\}. \quad (16)$$

To emphasize high-frequency components associated with boundaries and fine structures, we introduce a radial weighting function:

$$W(u, v) = 1 + \lambda \frac{\sqrt{(u - u_c)^2 + (v - v_c)^2}}{\sqrt{u_c^2 + v_c^2} + \epsilon}. \quad (17)$$

The final frequency loss is

$$\mathcal{L}_{\text{freq}} = \frac{1}{N} \sum_{u,v} W(u, v) |M_{\hat{D}}(u, v) - M_D(u, v)|^p. \quad (18)$$

The weight increases with the distance from the spectral center, assigning greater importance to high-frequency discrepancies while retaining supervision over the low-frequency structure. Applying this objective only after spatial convergence prevents premature amplification of local noise. The final phase therefore focuses on recovering thin structures and sharp depth discontinuities without introducing additional frequency-domain computation during inference.

TABLE I  
ZERO-SHOT DEPTH ESTIMATION. THE BEST AND SECOND-BEST RESULTS AMONG ALL METHODS ARE SHOWN IN BOLD AND UNDERLINED, RESPECTIVELY.

| Type           | Method                | Training Data | NYUv2      |                     | KITTI      |                     | ETH3D      |                     | ScanNet    |                     | DIODE      |                     |
|----------------|-----------------------|---------------|------------|---------------------|------------|---------------------|------------|---------------------|------------|---------------------|------------|---------------------|
|                |                       |               | AbsRel↓    | $\delta_1 \uparrow$ | AbsRel↓    | $\delta_1 \uparrow$ | AbsRel↓    | $\delta_1 \uparrow$ | AbsRel↓    | $\delta_1 \uparrow$ | AbsRel↓    | $\delta_1 \uparrow$ |
| Discriminative | DiverseDepth [47]     | 320K          | 11.7       | 87.5                | 19.0       | 70.4                | 22.8       | 69.4                | 10.9       | 88.2                | –          | –                   |
|                | MiDaS [14]            | 2M            | 11.1       | 88.5                | 23.6       | 63.0                | 18.4       | 75.2                | 12.1       | 84.6                | –          | –                   |
|                | LeReS [15]            | 354K          | 9.0        | 91.6                | 14.9       | 78.4                | 17.1       | 77.7                | 9.1        | 91.7                | –          | –                   |
|                | Omnidata [48]         | 12.2M         | 7.4        | 94.5                | 14.9       | 83.5                | 16.6       | 77.8                | 7.5        | 93.6                | –          | –                   |
|                | HDN [49]              | 300K          | 6.9        | 94.8                | 11.5       | 86.7                | 12.1       | 83.3                | 8.0        | 93.9                | –          | –                   |
|                | DPT [13]              | 1.4M          | 9.8        | 90.3                | 10.0       | 90.1                | 7.8        | 94.6                | 8.2        | 93.4                | –          | –                   |
|                | Depth Anything V2 [1] | 63.5M         | <b>4.3</b> | <b>98.0</b>         | <b>8.0</b> | <b>94.6</b>         | 6.2        | <b>98.0</b>         | <b>4.3</b> | <b>98.1</b>         | <b>6.6</b> | <b>95.2</b>         |
| Generative     | Marigold [4]          | 74K           | 5.5        | 96.4                | 9.9        | 91.6                | 6.5        | 96.0                | 6.4        | 95.1                | 10.0       | 90.7                |
|                | GeoWizard [50]        | 280K          | 5.2        | 96.6                | 9.7        | 92.1                | 6.4        | 96.1                | 6.1        | 95.3                | 12.0       | 89.8                |
|                | DepthFM [44]          | 74K           | 6.0        | 95.5                | 9.1        | 90.2                | 6.5        | 95.4                | 6.6        | 94.9                | –          | –                   |
|                | GenPercept [5]        | 90K           | 5.6        | 96.0                | 9.9        | 90.4                | 6.2        | 95.8                | 5.6        | 96.5                | –          | –                   |
|                | Lotus [6]             | 54K           | 5.3        | 96.7                | 8.5        | 92.8                | 5.9        | 97.0                | 5.9        | 95.7                | 9.8        | 92.4                |
|                | DepthMaster [45]      | 74K           | 5.0        | 97.2                | 8.2        | 93.7                | <b>5.3</b> | <b>97.4</b>         | 5.5        | 96.7                | –          | –                   |
|                | SCFDepth (Ours)       | 74K           | <u>4.5</u> | <b>97.3</b>         | 8.7        | 93.7                | <u>5.6</u> | <u>96.3</u>         | <u>4.5</u> | <u>97.3</u>         | <u>8.3</u> | <u>93.2</u>         |



Fig. 3. Frequency-domain refinement sharpens high-frequency depth boundaries after the global structure has converged.

## IV. EXPERIMENTS

### A. Implementation Details

SCFDepth is built upon Stable Diffusion v2 [24]. We disable text cross-attention and set the prediction type to `sample` for deterministic single-step inference. Stage 1 is trained for 20K iterations, while Stage 2 is trained for 21K iterations, including 20K iterations of spatial supervision and 1K iterations of frequency refinement. Both stages use a batch size of 32 on a single NVIDIA RTX 6000 Ada Generation GPU and require approximately 7.7 and 10 days, respectively. Evaluation is performed on a single NVIDIA GeForce RTX 4090, requiring approximately 30 minutes for all benchmarks.

The two stages sequentially optimize the same U-Net rather than forming an inference-time cascade. The external encoder is used only for feature alignment in Stage 1 and is removed afterward. Stage 2 initializes from the aligned U-Net and introduces the RGB and pretrained-MDE prior latents for supervised depth learning. During inference, all auxiliary losses and frequency-domain operations are removed. The deployed model consists of the selected depth prior, VAE encoding, one U-Net prediction, and VAE decoding, without iterative sampling or test-time ensembling.

### B. Datasets and Evaluation Protocol

**Training Datasets.** We train SCFDepth exclusively on synthetic data from Hypersim [51] and Virtual KITTI [52]. Following the Marigold split, approximately 54K Hypersim images are used at a resolution of  $480 \times 640$ . Virtual

KITTI contributes approximately 20K road-scene images at  $1216 \times 352$  resolution under diverse weather and illumination conditions, with the far plane fixed at 80 m. The two datasets are sampled at a ratio of 9:1, and no image from the evaluation datasets is used for training.

**Evaluation Datasets.** We conduct zero-shot evaluation on five real-world benchmarks. We use the official 654-image test split of NYUv2 [53], the 800-image ScanNet split adopted by Marigold [54], and the 652-image KITTI Eigen split [7], [55]. ETH3D and DIODE are evaluated using the Marigold protocol with 454 and 771 images, respectively [56], [57]. These datasets cover indoor scenes, road environments, and mixed indoor-outdoor geometry.

**Evaluation Protocol.** Following affine-invariant evaluation [14], each predicted depth map  $\hat{m}$  is aligned to the ground-truth depth  $d$  using least-squares scale and shift fitting:

$$\tilde{d} = a\hat{m} + b. \quad (19)$$

We report Absolute Mean Relative Error,

$$\text{AbsRel} = \frac{1}{M} \sum_{i=1}^M \frac{|\tilde{d}_i - d_i|}{d_i}, \quad (20)$$

and threshold accuracy  $\delta_1$ , defined as the proportion of valid pixels satisfying

$$\max\left(\frac{\tilde{d}_i}{d_i}, \frac{d_i}{\tilde{d}_i}\right) < 1.25. \quad (21)$$

The scale and shift parameters are fitted independently for each test image. For DIODE, we exclude invalid values and depths outside 0.6–350 m. Pixels whose ground-truth Sobel gradient magnitude exceeds 0.3 are additionally removed to reduce the influence of severe boundary misalignment. The same mask is applied to all compared predictions, and the final valid region is the intersection of the dataset-valid mask, depth-range mask, and gradient mask.

### C. Ablation Studies

We evaluate the effects of the pretrained depth prior, feature adapter, frequency weighting, external encoder, and training-data composition while keeping all other settings unchanged.



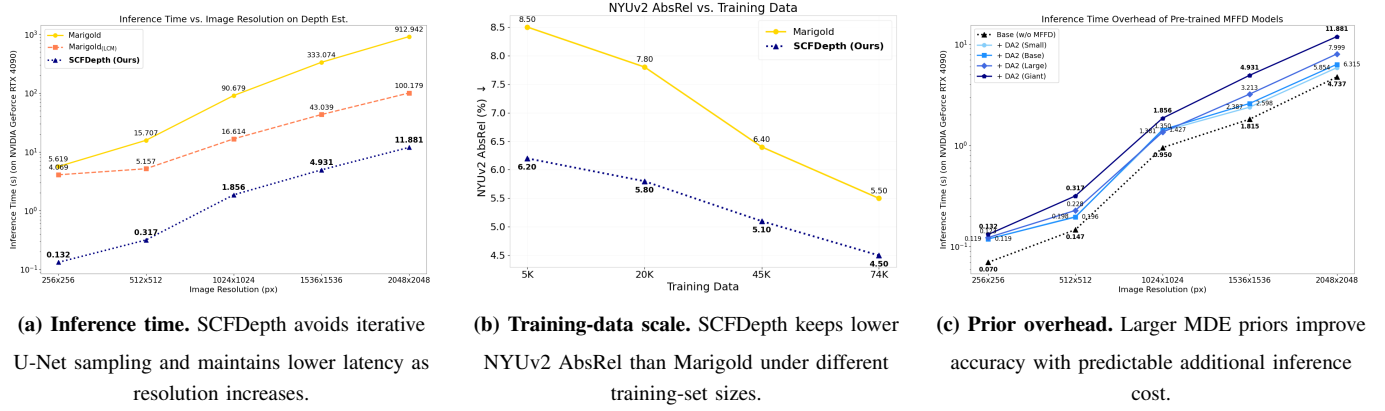


Fig. 4. Efficiency analysis of SCFDepth. The left plot compares inference time across image resolutions, the middle plot reports NYUv2 AbsRel under different training-data scales, and the right plot shows the additional inference overhead caused by pretrained MDE priors with different capacities.

TABLE II  
ABLATION OF THE PRETRAINED MDE PRIOR.

| Prior     | NYUv2      |                     | KITTI      |                     |
|-----------|------------|---------------------|------------|---------------------|
|           | AbsRel↓    | $\delta_1 \uparrow$ | AbsRel↓    | $\delta_1 \uparrow$ |
| base      | 5.3        | 96.5                | 10.7       | 89.3                |
| DA2 small | 5.2        | 96.7                | 9.7        | 92.1                |
| DA2 base  | 5.1        | 96.8                | 9.2        | 92.6                |
| DA2 large | 4.7        | 97.1                | 9.1        | 93.2                |
| DA2 giant | <b>4.5</b> | <b>97.1</b>         | <b>8.7</b> | <b>93.7</b>         |

**Pretrained MDE Prior.** As shown in Table II, all DA2 variants improve upon the baseline, and performance generally increases with prior capacity. Even DA2-small reduces the KITTI AbsRel from 10.7 to 9.7, indicating that a lightweight prior already provides useful geometric guidance. DA2-giant achieves the best accuracy, while Fig. 4(c) shows the corresponding increase in inference cost, allowing different accuracy–efficiency configurations.

TABLE III  
ABLATION OF STAGE-1 ADAPTER TYPE.

| Adapter      | NYUv2      |                     | KITTI      |                     |
|--------------|------------|---------------------|------------|---------------------|
|              | AbsRel↓    | $\delta_1 \uparrow$ | AbsRel↓    | $\delta_1 \uparrow$ |
| MLP Adapter  | 4.7        | 96.7                | 8.9        | 93.3                |
| Conv Adapter | <b>4.5</b> | <b>97.1</b>         | <b>8.7</b> | <b>93.7</b>         |

**Adapter Architecture.** Table III shows that the Conv Adapter consistently outperforms the MLP Adapter on both NYUv2 and KITTI. This supports preserving local spatial context when aligning intermediate U-Net features with external representations.

TABLE IV  
ABLATION OF HIGH-PASS WEIGHT  $\lambda$ .

| Setting         | NYUv2      |                     | KITTI      |                     |
|-----------------|------------|---------------------|------------|---------------------|
|                 | AbsRel↓    | $\delta_1 \uparrow$ | AbsRel↓    | $\delta_1 \uparrow$ |
| $\lambda = 0.6$ | <b>4.5</b> | <b>97.1</b>         | <b>8.7</b> | <b>93.7</b>         |
| $\lambda = 1.0$ | 4.5        | 97.1                | 8.8        | 93.7                |
| $\lambda = 1.6$ | 4.6        | 97.0                | 8.8        | 93.5                |
| $\lambda = 2.0$ | 4.6        | 97.0                | 8.8        | 93.6                |

**High-Pass Weight.** As shown in Table IV,  $\lambda = 0.6$  provides the best overall balance. Increasing the weight does not further

improve the depth metrics and slightly degrades global accuracy, suggesting that excessive emphasis on high-frequency components may disturb the established depth structure.

TABLE V  
ABLATION OF EXTERNAL ENCODERS.

| Encoder             | NYUv2      |                     | KITTI      |                     |
|---------------------|------------|---------------------|------------|---------------------|
|                     | AbsRel↓    | $\delta_1 \uparrow$ | AbsRel↓    | $\delta_1 \uparrow$ |
| Baseline (w/o Ext.) | 4.5        | 96.8                | 8.8        | 93.5                |
| DINOv2 [46]         | <b>4.5</b> | <b>97.1</b>         | <b>8.7</b> | <b>93.7</b>         |
| OpenCLIP [58]       | 4.7        | 96.8                | 8.7        | 93.7                |
| SAM [59]            | 4.6        | 97.0                | 8.8        | 93.5                |

**External Encoder.** Table V compares different external encoders. DINOv2 achieves the most consistent performance across NYUv2 and KITTI, whereas OpenCLIP and SAM provide limited or inconsistent improvements. This result indicates that the localized patch representations of DINOv2 are better suited to dense feature alignment in SCFDepth.

TABLE VI  
ABLATION OF TRAINING DATASETS.

| Hypersim | V-KITTI | NYUv2      |                     | KITTI      |                     |
|----------|---------|------------|---------------------|------------|---------------------|
|          |         | AbsRel↓    | $\delta_1 \uparrow$ | AbsRel↓    | $\delta_1 \uparrow$ |
| ✗        | ✓       | 10.3       | 90.9                | 10.4       | 90.3                |
| ✓        | ✗       | 4.7        | 96.6                | 9.2        | 92.1                |
| ✓        | ✓       | <b>4.5</b> | <b>97.1</b>         | <b>8.4</b> | <b>93.1</b>         |

**Training Data.** Table VI shows that training on Hypersim produces substantially better indoor performance than using Virtual KITTI alone. Adding Virtual KITTI further improves KITTI performance while maintaining the NYUv2 results, demonstrating the benefit of combining indoor and outdoor synthetic scenes.

#### D. Main Results Analysis

Table I summarizes the zero-shot performance of SCFDepth on five real-world benchmarks. Compared with multi-step diffusion methods, SCFDepth maintains competitive geometric accuracy while using only one denoising step. On NYUv2 and ScanNet, the proposed method achieves strong indoor performance, indicating that the pretrained geometric prior and

feature alignment effectively preserve global scene structure. On KITTI and ETH3D, SCFDepth also remains competitive with recent generative models, showing that the single-step formulation does not rely solely on indoor scene statistics.

The efficiency analysis in Fig. 4 further illustrates the practical advantage of SCFDepth. As the image resolution increases, iterative diffusion models suffer from rapidly growing inference time because the U-Net must be repeatedly evaluated. In contrast, SCFDepth performs a single deterministic prediction, leading to a substantially lower runtime at all tested resolutions. The training-data comparison also shows that SCFDepth benefits more consistently from limited data, suggesting that the external geometric prior and the two-stage curriculum reduce the burden of learning the complete depth distribution from scratch.

The right part of Fig. 4 reports the overhead introduced by different pretrained MDE priors. Larger priors improve accuracy but also increase inference cost, providing a flexible accuracy–efficiency trade-off. This behavior is important for deployment: lightweight priors can be selected for latency-sensitive scenarios, while stronger priors can be used when peak accuracy is preferred.

## V. CONCLUSION

SCFDepth combines deterministic one-step diffusion, a feed-forward geometric prior, spatial-preserving feature alignment, and scheduled frequency refinement. The method avoids iterative sampling while retaining coherent global depth and sharp local boundaries, achieving strong zero-shot results with modest training data. The ablations verify that each component contributes to the final accuracy–efficiency balance.

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